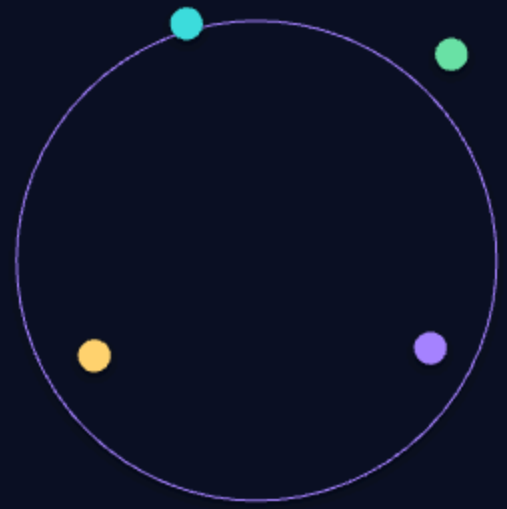
Quantum Campaign Optimizer

A transparent three-stage quantum–classical marketing pipeline

Quantum campaign optimization

Four-qubit learning → ten-qubit selection → six-qubit simulation

Shylan Kamesh S
Sai Santhosh T
Derin Godson



Atomiq 2026

Bank term-deposit campaign

What the shared Atomiq problem asks us to compute

01 Who will convert?

Predict probabilities for all 100 unseen customers; report quality and rank the top ten.

02 Who should we target?

Select exactly six from ten under official budget and capacity limits.

03 Will we hit the target?

Estimate the probability that at least three of those six convert.

300 labeled training customers, 100 hidden-label test customers. Duration excluded.

End-to-end architecture

Unbroken customer-ID and continuous-probability traceability

INPUT

300 training + 100 test customers

›

STAGE 1

4Q QML kernel + classical benchmarks

›

STAGE 2

10Q XY-QAOA + 210 exact checks

›

STAGE 3

6Q RY encoding, shots and noise

›

OUTPUT

Six IDs and campaign-success probability

Guardrails: training-only preprocessing • measured QAOA gap • exact analytical Stage 3 check

Stage 1 / Four-qubit quantum learning

Executed FidelityQuantumKernel + QSVC, three-fold training-only validation

Quantum feature map

RY(πx) + RZ(πx) on each qubit

Neighbor RZZ entanglers

Quantum kernel: $|\langle\phi(x)|\phi(y)\rangle|^2$

SVC probability model

0.163

OOF average precision

0.100

OOF Brier score

100%

Quantum blend share

Training-only model selection

- Fit transforms and feature selection only on training fold
- Freeze OOF selection, then refit. Quantum subset: 64 rows.

Stage 2 / Ten-qubit constrained optimization

QAOA candidate sampling + an exact classical feasibility certificate

210

Exact six-person combinations

0.000

Observed QAOA objective gap

6 / 10

Selected from candidates

Maximize $\Sigma\, x_i(p_i V_i - c_i)$, $\Sigma\, x_i = 6$

Budget 9/25, capacity 19/22, net value 6.1375

XY mixer preserves exactly six (not budget). Exact solver certifies final IDs; no speedup claim.

Final customer IDs

C0005
C0007
C0009
C0020
C0022
C0036

Stage 3 / Six-qubit conversion simulation

An interpretable, independently benchmarked probability

$$\theta_i = 2 \operatorname{asin}(\sqrt{p_i})$$

$RY(\theta_0) \bullet RY(\theta_1) \bullet \dots RY(\theta_5)$

Measure six qubits

Success ⇢ at least three “1” outcomes

0.022797

Ideal sampled tail probability

0.024049

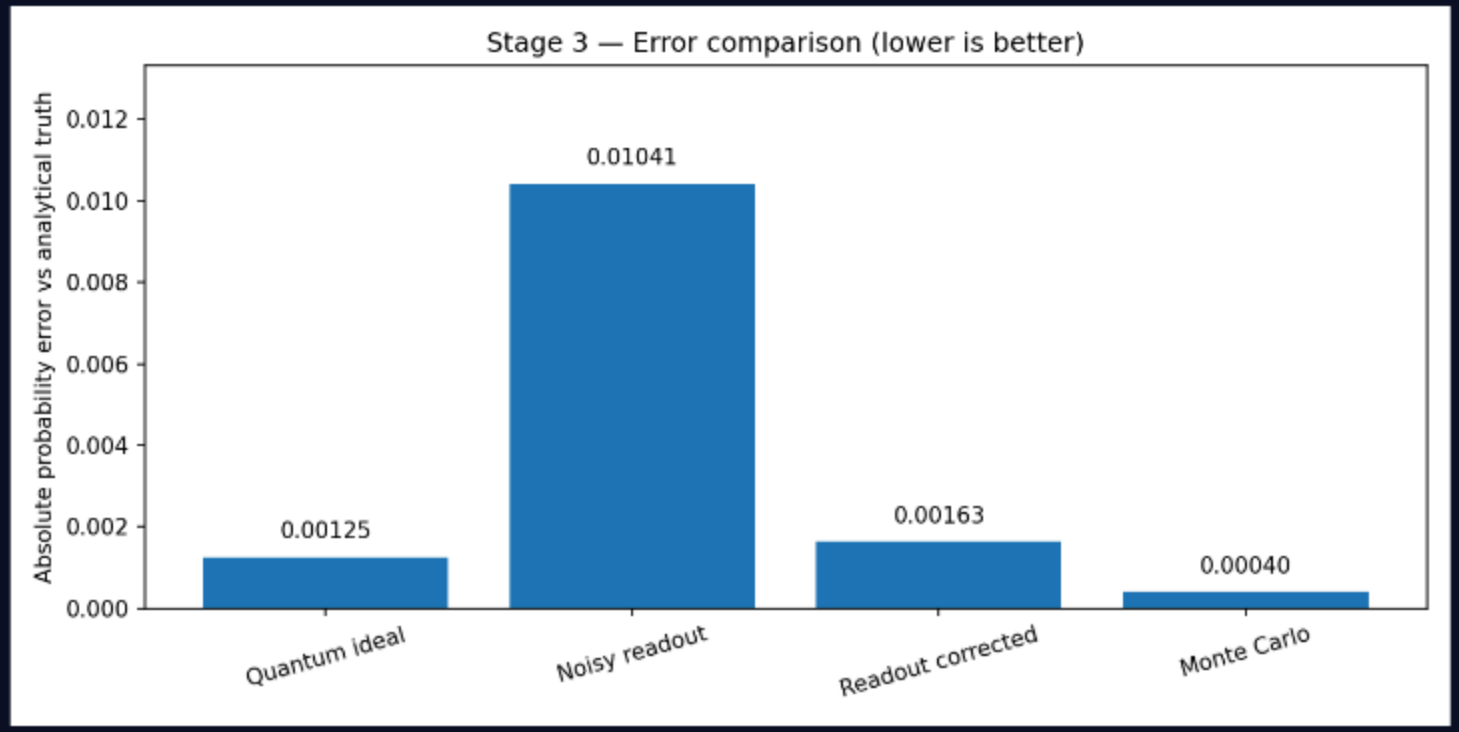
Exact Poisson–binomial reference

32,768 Aer shots. Depth 2. Rightmost bit = q0.

95% shot interval: 0.021236–0.024469.

Noise and error measurements

Independent 2.5% readout flips; exact known-matrix correction



Observed results

- Ideal: 0.022797; exact: 0.024049
- Noisy: 0.034454
- Mitigated: 0.022420

Synthetic independent readout model ≠ device-calibrated noise.

Quantum and classical computation

A precise, reproducible research boundary

Quantum components

4Q FidelityQuantumKernel + QSVC
10Q XY-QAOA, 8,192 Aer shots
6Q RY simulation, 32,768 Aer shots
Qiskit 2.6.2, Aer 0.17.2, QML 0.9.1

Classical components

Data preprocessing and model fitting
Optimizer parameter search
Exact enumeration of 210 subsets
Exact Poisson-binomial + Monte Carlo

QAOA depth: 44 Aer-native; 6,144 in u/cx basis. No hardware or speedup claim.

Prediction uncertainty and limitations

Training OOF metrics guide selection; hidden test accuracy is unknown

01 QSVC has OOF average precision 0.163 and ROC-AUC 0.549

02 Four selected categorical indicators give 86 tied top probabilities

03 Official alphanumeric tie rule determines the top-ten candidate pool

04 200 probability perturbations: mean overlap 0.67 of six selected IDs

05 Sensitivity is illustrative, not a confidence interval or evidence of advantage

Why the XY mixer matters

Cardinality is guaranteed by the ansatz; budget/capacity still need checks

XY-QAOA / 10 qubits

Uniform weight-six initial state

XY ring preserves cardinality

Measured feasible fraction: 100%

All six XY trials measured the optimum

X-mixer / original baseline

Plus state over all 1,024 states

RX can change the selection count

Depth 1 mean feasibility: 31.6%

Depth 2 mean feasibility: 26.0%

Seeds 11, 22, 33; depths 1, 2; 2,048 shots. All 210 six-person subsets are feasible here.

Campaign result and reproducibility

Expected net value 6.1375; quantum probability of at least three conversions 2.28%

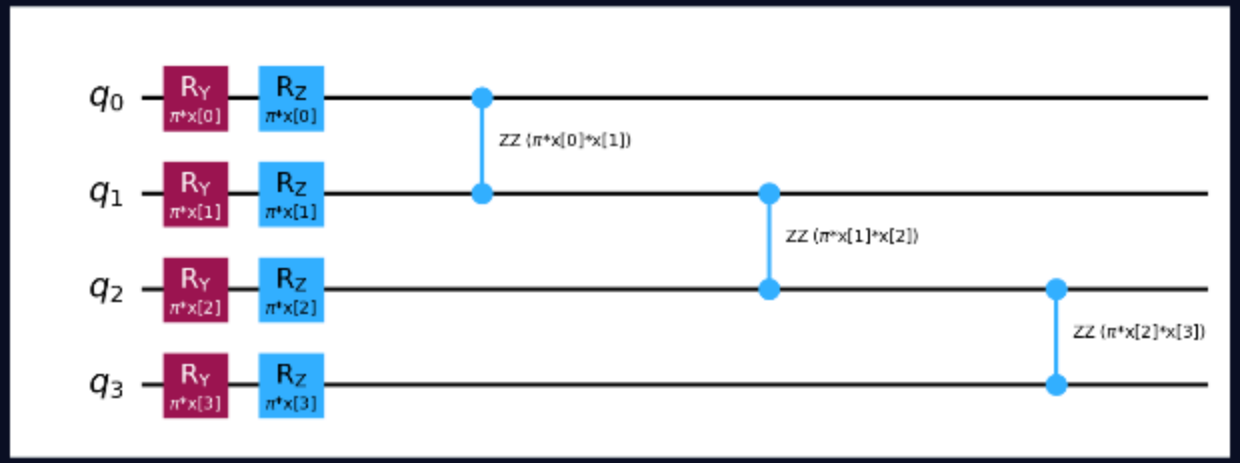
4 qubits → 10 qubits → 6 qubits

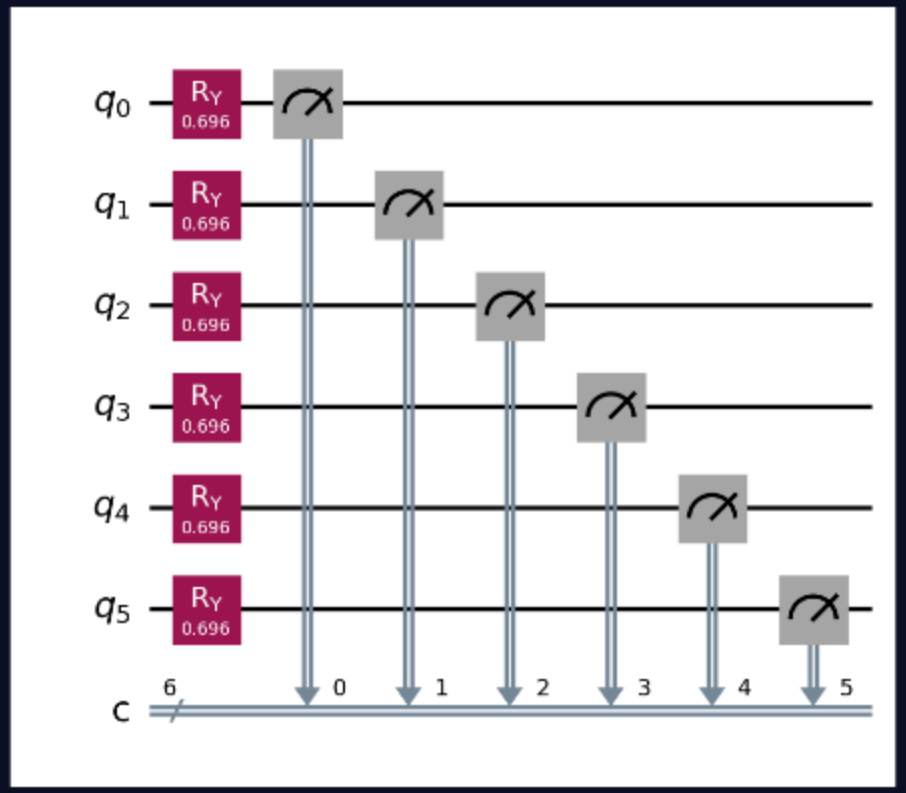
44 tests passed, zero skipped. Independent output audit passed. Exact dependencies, seeds, source hashes and measurement counts included.

github.com/shylankamesh/atomiq-quantum-campaign-2026

Executed quantum circuit designs

4-qubit feature map and 6-qubit campaign circuit





Feature map: logical depth 5, u/cx depth 11. Campaign: depth 2, six RY gates, no entanglers.

Qiskit-rendered diagrams. Simulator execution only. q0 is the rightmost measured bit.